

NETWORK SERVER DISCOVERY & HTTP SERVICE ANALYSIS USING NMAP

Project Portfolio Report

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2. Objective

The objective of this project was to build a controlled local laboratory environment and investigate how a network service can be discovered and analyzed using common Linux and cybersecurity tools.

3. Lab Environment

Component.	Detail
Operating System	Kali Linux
Virtualization	VirtualBox
Network interface	eth0
Kali IP	10.0.2.15
Network	10.0.2.0/24
Service	Python SimpleHTTPServer
Port	TCP 8000
Tools	Nmap, curl, ss

4. Methodology

I performed the project in several stages:

- Stage 1 — Network identification

I used:

`ip addr`

`ip route`

to identify the Kali VM's network interface, IP address, and local network.

- Stage 2 — Creating a controlled HTTP service

I created a small test webpage and launched a Python HTTP server on port 8000.

Initially, the service was bound to:

127.0.0.1:8000

This restricted access to the local machine.

- Stage 3 — Service discovery

I used Nmap to identify the service:

`nmap -sV -p 8000 127.0.0.1`

Nmap identified:

8000/tcp open http SimpleHTTPServer 0.6

- Stage 4 — Network binding

The server was then bound to the Kali network interface: 10.0.2.15:8000

We verified the listening address using ss.

- Stage 5 — Network service scanning

I scanned the network-interface address:

`nmap -sV -p 8000 10.0.2.15`

The service was again identified as an open HTTP service.

- Stage 6 — HTTP analysis.

I used Nmap's HTTP title script and curl to inspect the service.

The HTTP response included:

HTTP/1.0 200 OK

Server: SimpleHTTP/0.6 Python/...

Content-type: text/html

5. Key Findings

- Finding 1 — Open port

TCP port 8000 was open because the Python HTTP server was actively listening on it.

- Finding 2 — Service identification

Nmap successfully identified the service as SimpleHTTPServer 0.6.

- Finding 3 — Service binding

Changing the listening address from 127.0.0.1 to 10.0.2.15 demonstrated how the interface a service binds to affects its potential network exposure.

→ Finding 4 — Information disclosure

The HTTP Server header disclosed information about the software serving the webpage.

6. What I Learned

This project helped me understand that network reconnaissance is more than simply running a scanning command. I learned how to:

- Identify network interfaces and IP addresses.
- Create a controlled HTTP service.
- Understand TCP ports and listening services.
- Use Nmap for service discovery.
- Use ss to identify listening services and processes.
- Use curl to inspect HTTP responses.
- Understand the difference between localhost and network-interface binding.
- Recognize basic information disclosure through HTTP headers.

7. Security Consideration

The experiment was performed entirely in a controlled environment on my own Kali Linux virtual machine. No third-party systems or unauthorized networks were scanned.

8. Conclusion

This project provided practical experience with basic network reconnaissance and HTTP service analysis. It strengthened my understanding of how services listen on network interfaces, how security tools identify those services, and how seemingly small configuration decisions can affect network exposure.

1.

```

kali@kali:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86299sec preferred_lft 86299sec
    inet6 fd17:625c:f037:2:1c0:bcd6:b0d6:22cf/64 scope global dynamic noprefi
        valid_lft 86300sec preferred_lft 14300sec
    inet6 fe80::bda2:c32e:f1a6:3d5d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ip route
default via 10.0.2.1 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100

```

Figure 1.1: Kali Linux network configuration.

2.

```

kali@kali:~$ cd ~/cyber-lab
kali@kali:~/cyber-lab$ echo "Cybersecurity Lab Target" > index.html
kali@kali:~/cyber-lab$ ls
index.html
kali@kali:~/cyber-lab$ python3 -m http.server 8000 --bind 127.0.0.1
Serving HTTP on 127.0.0.1 port 8000 (http://127.0.0.1:8000/) ...
127.0.0.1 - - [16/Aug/2026 10:25:20] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [16/Aug/2026 10:25:20] code 501, message Unsupported method ('POST')
127.0.0.1 - - [16/Aug/2026 10:25:20] "POST /sdk HTTP/1.1" 501 -
127.0.0.1 - - [16/Aug/2026 10:25:20] code 404, message File not found
127.0.0.1 - - [16/Aug/2026 10:25:20] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [16/Aug/2026 10:25:20] "GET /nmaplowercheck1786890320 HTTP/1.1" 404 -
127.0.0.1 - - [16/Aug/2026 10:25:20] code 404, message File not found
127.0.0.1 - - [16/Aug/2026 10:25:20] "GET /evox/about HTTP/1.1" 404 -
127.0.0.1 - - [16/Aug/2026 10:25:20] code 404, message File not found
127.0.0.1 - - [16/Aug/2026 10:25:20] "GET /HNAPl HTTP/1.1" 404 -
127.0.0.1 - - [16/Aug/2026 10:25:20] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [16/Aug/2026 10:25:20] "GET / HTTP/1.1" 200 -

```

Figure 2.1: Creating HTTP service

3.

```
(kali㉿kali)-[~]
$ nmap -sV -p 8000 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-16 10:25 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).

PORT      STATE SERVICE VERSION
8000/tcp  open  http   SimpleHTTPServer 0.6 (Python 3.14.6)

Service detection performed. Please report any incorrect results at https://nmap.org/s
Nmap done: 1 IP address (1 host up) scanned in 8.79 seconds
(kali㉿kali)-[~]
$
```

Figure 3.1: Local Nmap scan

4.

```
(kali㉿kali)-[~/cyber-lab]
$ python3 -m http.server 8000 --bind 10.0.2.15
Serving HTTP on 10.0.2.15 port 8000 (http://10.0.2.15:8000/) ...
10.0.2.15 - - [16/Aug/2026 11:37:14] "GET / HTTP/1.0" 200 -
10.0.2.15 - - [16/Aug/2026 11:37:14] "GET / HTTP/1.0" 200 -
10.0.2.15 - - [16/Aug/2026 11:37:14] code 404, message File not found
10.0.2.15 - - [16/Aug/2026 11:37:14] "GET /nmaplowercheck1786894634 HTTP/1.1" 404 -
10.0.2.15 - - [16/Aug/2026 11:37:14] "GET / HTTP/1.1" 200 -
10.0.2.15 - - [16/Aug/2026 11:37:14] code 501, message Unsupported method ('POST')
10.0.2.15 - - [16/Aug/2026 11:37:14] "POST /sdk HTTP/1.1" 501 -
10.0.2.15 - - [16/Aug/2026 11:37:14] code 404, message File not found
10.0.2.15 - - [16/Aug/2026 11:37:14] "GET /HNAP1 HTTP/1.1" 404 -
10.0.2.15 - - [16/Aug/2026 11:37:14] code 404, message File not found
10.0.2.15 - - [16/Aug/2026 11:37:14] "GET /evox/about HTTP/1.1" 404 -
10.0.2.15 - - [16/Aug/2026 11:37:14] "GET / HTTP/1.0" 200 -
10.0.2.15 - - [16/Aug/2026 11:37:14] "GET / HTTP/1.1" 200 -
10.0.2.15 - - [16/Aug/2026 11:42:34] "HEAD / HTTP/1.1" 200 -
```

Figure 4.1: Network Nmap scan

5.

```
(kali@kali)-[~]
└─$ curl http://127.0.0.1:8000
Cybersecurity Lab Target

(kali@kali)-[~]
└─$ ss -ltnp | grep 8000
LISTEN 0      5            10.0.2.15:8000      0.0.0.0:*        users:(("python3",pid=18138,fd=3))

(kali@kali)-[~]
└─$ nmap -sV -p 8000 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-16 10:50 -0400
Nmap scan report for 10.0.2.15
Host is up (0.00024s latency).

PORT      STATE SERVICE VERSION
8000/tcp  open  http      SimpleHTTPServer 0.6 (Python 3.14.6)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 12.63 seconds

(kali@kali)-[~]
```

Figure 5.1: Network service scanning

6.

```
(kali@kali)-[~]
└─$ nmap -sV --script=http-title -p 8000 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-16 11:37 -0400
Nmap scan report for 10.0.2.15
Host is up (0.00019s latency).

PORT      STATE SERVICE VERSION
8000/tcp  open  http      SimpleHTTPServer 0.6 (Python 3.14.6)
|_http-title: Site doesn't have a title (text/html).
|_http-server-header: SimpleHTTP/0.6 Python/3.14.6

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 12.73 seconds

(kali@kali)-[~]
└─$ curl -I http://10.0.2.15:8000
HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.14.6
Date: Sun, 16 Aug 2026 15:42:34 GMT
Content-type: text/html
Content-Length: 25
Last-Modified: Sun, 16 Aug 2026 14:21:31 GMT

(kali@kali)-[~]
```

Figure 6.1: HTTP analysis